

Temporal Normalization

Query Side



Query

"What book was I reading during my Europe trip last month"



$Norm(q, t_q)$

Normalized query $\tilde{q}$

"What book was I reading during my Europe trip last month (Mar, 2026)?"
 $H_0 = \emptyset$

Memory Side

Session 3 — Apr 2

"Last week, my Europe trip was fun; I visited Paris and Rome."

Session 2 — Mar 25

"I just finished reading The Lord of the Rings yesterday."



$Norm(S, t)$

Normalized Session 3

"...Last week (Mar 23-29), my Europe trip was fun;

Normalized Session 2

"...reading The Lord of the Rings yesterday (Mar 24)."

Evidence-Driven Search Loop



Planner

Round 1: $\tilde{q} + C_{1,2}$

Cue 1: "When did I trip to Europe last month (Mar, 2026)"
 Cue 2: "What book had I read?"

Round 2: C_3

Cue 3: "What book was I reading during Mar 23-29?"



Hybrid Retriever



Normalized Session 3:
"Last week (Mar 23-29), my Europe trip was fun

Normalized Session 29:
 "I had a cook book..."



Round 1: Contains distractors



Normalized Session 2:
 "... reading The Lord of the Rings yesterday (Mar 24)."



Round 2: Use Temporal Hints



Synthesizer

Round 1: Reasoning (g_1):

"Found Europe trip event in Session 3 but no reading found in that time."

Evidence Events (E_1):

E1: ("Europe trip", $t_{\text{event}} = \text{Mar 23-29 from Session 3}$, $t_{\text{session}} = \text{Apr 2}$)

Verdict (v_1): Insufficient ❌

Round 2: Reasoning (g_2):

"Found reading The Lord of the Rings at Mar 24, in the week of Europe trip"

Evidence Events (E_2):

E2: ("Read The Lord of the Rings", $t_{\text{event}} = \text{Mar 24, from Session 2}$, $t_{\text{session}} = \text{Mar 25}$)

Verdict (v_2): Sufficient ✅

R1: Insufficient, Another round



Temporal Evidence Pool H_t

Round 1

E1: ("Europe trip", $t_{\text{event}} = \text{Mar 23-29 from Session 3}$, $t_{\text{session}} = \text{Apr 2}$)

Round 2

E2: ("Read The Lord of the Rings", $t_{\text{event}} = \text{Mar 24, from Session 2}$, $t_{\text{session}} = \text{Mar 25}$)

Append Events E_t From each round

Answer Model

R2: Sufficient, Finish